



TECHNO INSTITUTE OF ENGINEERING & MANAGEMENT BANIPPUR

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SMART ATTENDANCE SYSTEM

Submitted in Partial Fulfillment for the Degree of Bachelor of Technology

Submitted by:

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Introduction

- ❖ Title: Smart Attendance System using Face Recognition
 - ❖ One-line summary: A secure, automated attendance system that uses face recognition to accurately mark student attendance and reduce manual effort.
 - ❖ Scope: Real-time face detection & recognition, web-based admin dashboard, attendance logs and reports.
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Motivation

- ❖ Manual attendance is slow, error-prone, and easily falsified.
 - ❖ Large classes increase administrative burden and reduce teaching time.
 - ❖ COVID-era and hybrid classes heighten need for contactless, automated solutions.
 - ❖ Automated systems improve reliability, auditability and class analytics.
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Objectives

- ❖ Primary: Build a reliable face-based attendance system for classroom use.
- ❖ Technical objectives:
 1. Real-time face detection and recognition in browser.
 2. Secure backend to store attendance and manage users.
 3. Easy-to-use admin UI for class scheduling and reports.
- ❖ Success criteria: $\geq 90\%$ recognition accuracy in controlled classroom conditions; average check-in $< 5s$ per student.

Work Progress



❖ Completed:

1. Project scaffolding: React frontend (Vite) + Node/Express backend.
2. Face recognition integration in client using face-api.js.
3. Basic user models and attendance schema on backend.
4. Sample classes and seed data scripts.

❖ In progress:

1. Improving recognition accuracy and error handling.
 2. Implementing admin dashboard features (export, edit).
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Technology Required

Frontend	Backend	Dev & Ops	Database
React	Node.js	Git Hub	MongoDB
Vite	Express	Node	
Tailwind CSS	Mongoose		
Face-api.js			

Future Work Plans



- ❖ Short term (next 1–2 months):
 1. Improve accuracy (augmentation, more samples).
 2. Add admin exports (CSV/PDF) and attendance analytics.
- ❖ Medium term (3–6 months):
 1. Role-based access, multi-class scheduling, mobile-friendly UI.
 2. Integrate notifications (absent alerts).
- ❖ Long term:
 1. Multi-camera support, edge processing, federated privacy-preserving learning.
 2. Formal evaluation (user study) and deployment to campus.

References

Impact factor	Measurement	Target	Achieved
Audience interaction	Percentage (%)	85	88
Knowledge retention	Percentage (%)	75	80
Post-presentation surveys	Average rating	4.2	4.5
Referral rate	Percentage (%)	10	12
Collaboration opportunities	# of opportunities	8	10



Thank you

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